



Lecture 8

Module 3: AWS Global Infrastructure Overview

Module 4: AWS Cloud Security



Module 3: AWS Global Infrastructure Overview

AWS Academy Cloud Foundations



Module overview

Topics

- AWS Global Infrastructure
- AWS service and service category overview

Demo

- AWS Global Infrastructure

Activities

- AWS Management Console clickthrough



Knowledge check



Module objectives

After completing this module, you should be able to:

- Identify the difference between AWS Regions, Availability Zones, and edge locations
- Identify AWS service and service categories



Section 1: AWS Global Infrastructure

Module 3: AWS Global Infrastructure Overview



AWS Global Infrastructure

- The **AWS Global Infrastructure** is designed and built to deliver a **flexible, reliable, scalable**, and **secure** cloud computing environment with high-quality **global network performance**.
- AWS continually updates its global infrastructure footprint. Visit one of the following web pages for current infrastructure information:

- **AWS Global Infrastructure Map:**
https://aws.amazon.com/about-aws/global-infrastructure/#AWS_Global_Infrastructure_Map
Choose a circle on the map to view summary information about the Region represented by the circle.
- **Regions and Availability Zones:**
https://aws.amazon.com/about-aws/global-infrastructure/regions_az/
Choose a tab to view a map of the selected geography and a list of Regions, Edge locations, Local zones, and Regional Caches.

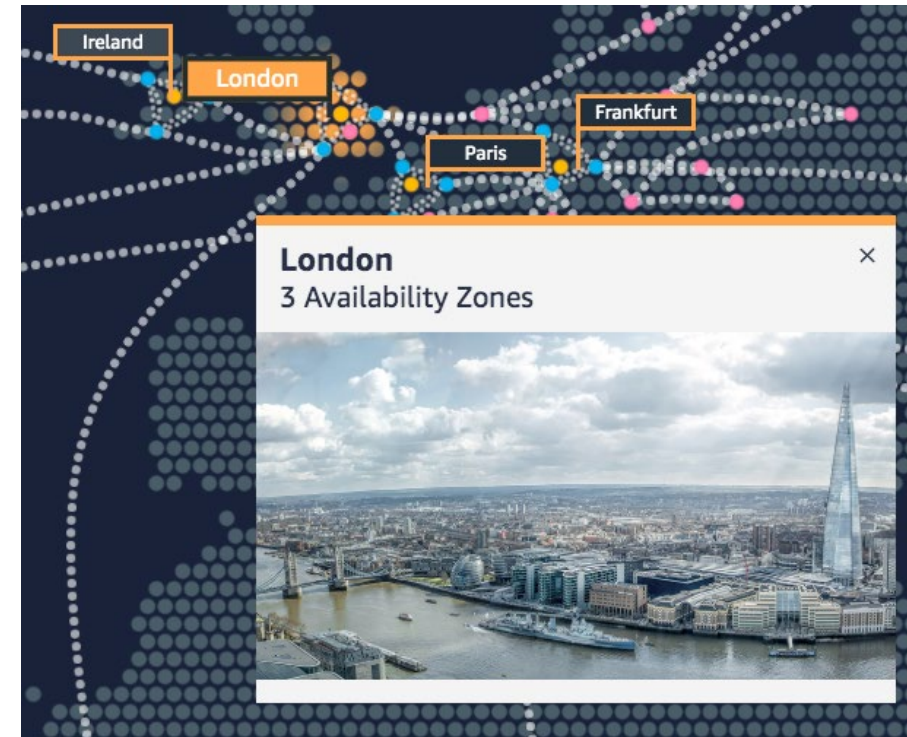


Educator-Led Demo: AWS Global Infrastructure Details



AWS Regions

- An **AWS Region** is a geographical area.
 - **Data replication** across Regions is controlled by you.
 - **Communication** between Regions uses AWS backbone network infrastructure.
- Each Region provides full redundancy and connectivity to the network.
- A Region typically consists of two or more **Availability Zones**.



Example: London Region



Selecting a Region

Determine the right Region for your services, applications, and data based on these factors



Data governance, legal requirements



Proximity to customers (latency)



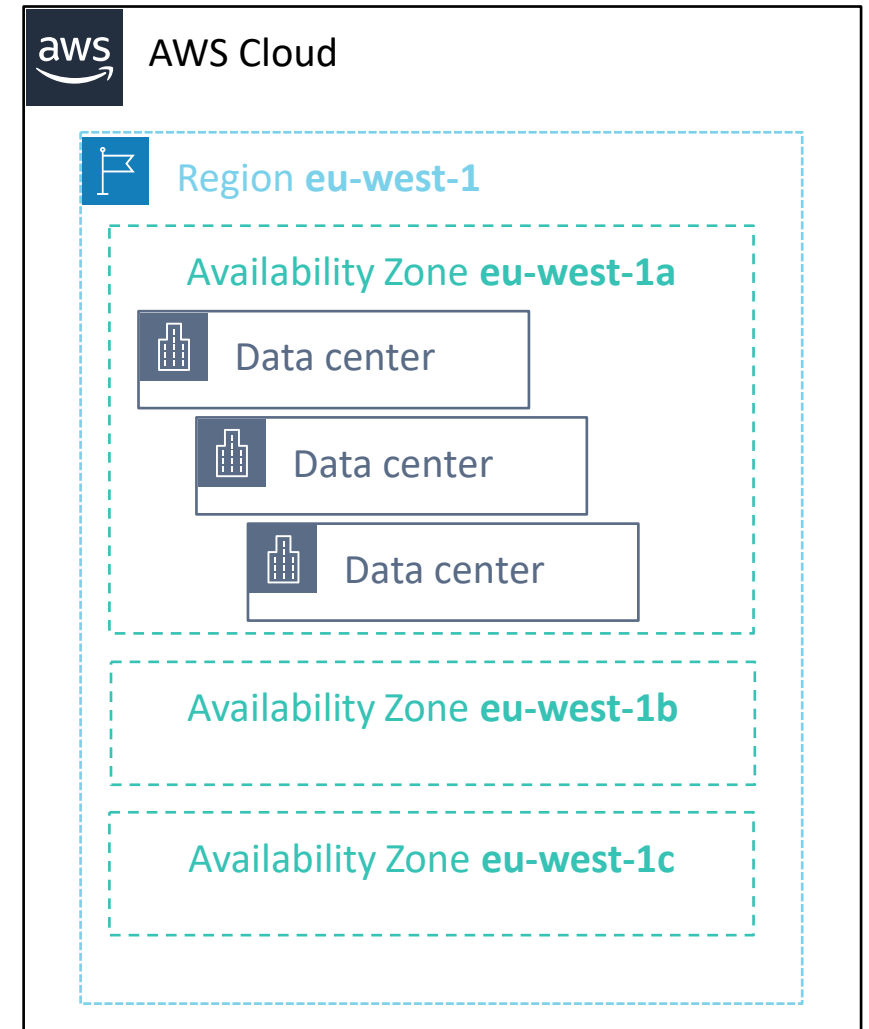
Services available within the Region



Costs (vary by Region)

Availability Zones

- Each **Region** has multiple Availability Zones.
- Each **Availability Zone** is a fully isolated partition of the AWS infrastructure.
 - Availability Zones consist of discrete **data centers**
 - They are designed for fault isolation
 - They are interconnected with other Availability Zones by using high-speed private networking
 - You choose your Availability Zones.
 - **AWS recommends replicating data and resources across Availability Zones** for resiliency.





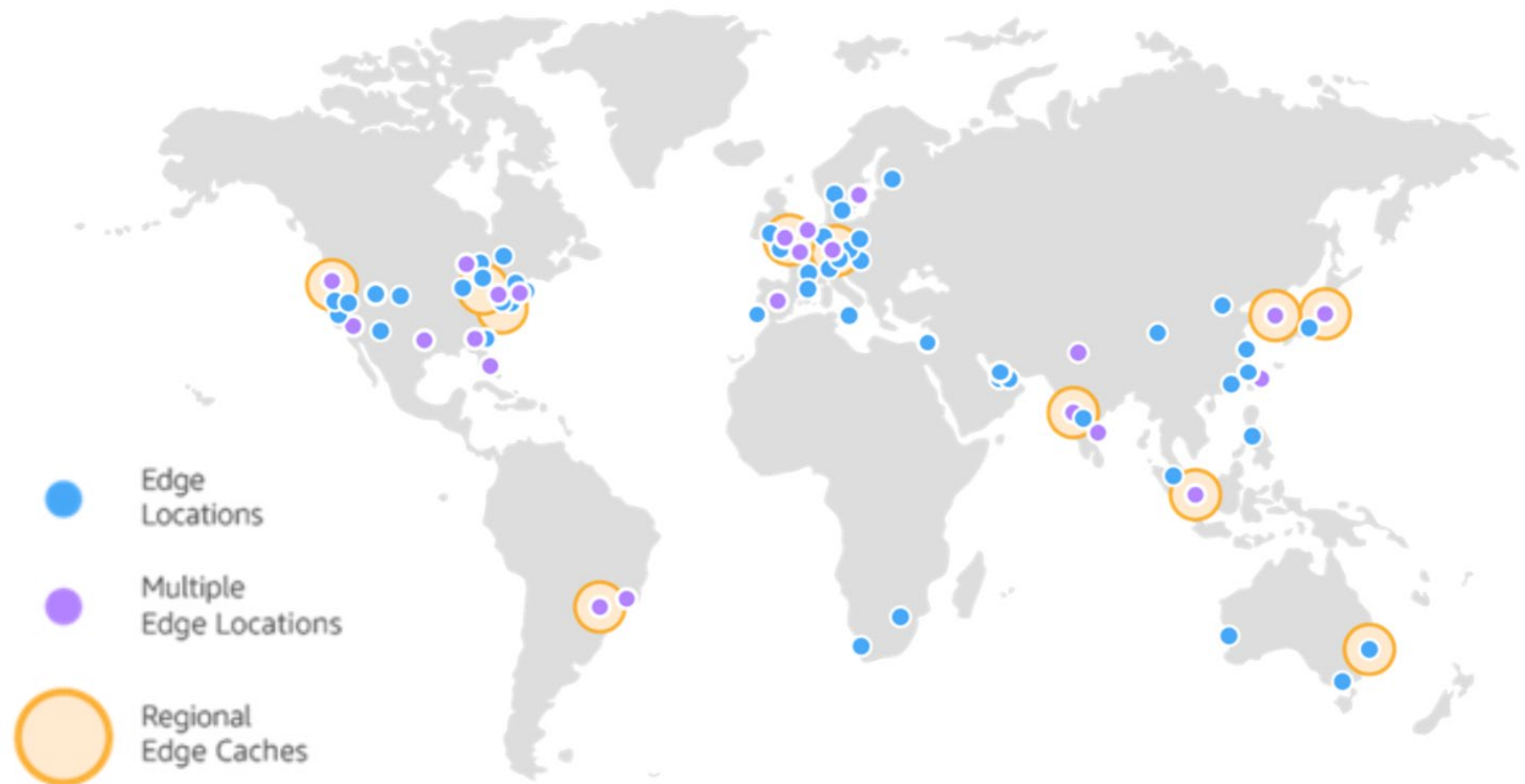
AWS data centers

- AWS data centers are designed for security.
- Data centers are where the data resides and data processing occurs.
- Each data center has redundant power, networking, and connectivity, and is housed in a separate facility.
- A data center typically has 50,000 to 80,000 physical servers.



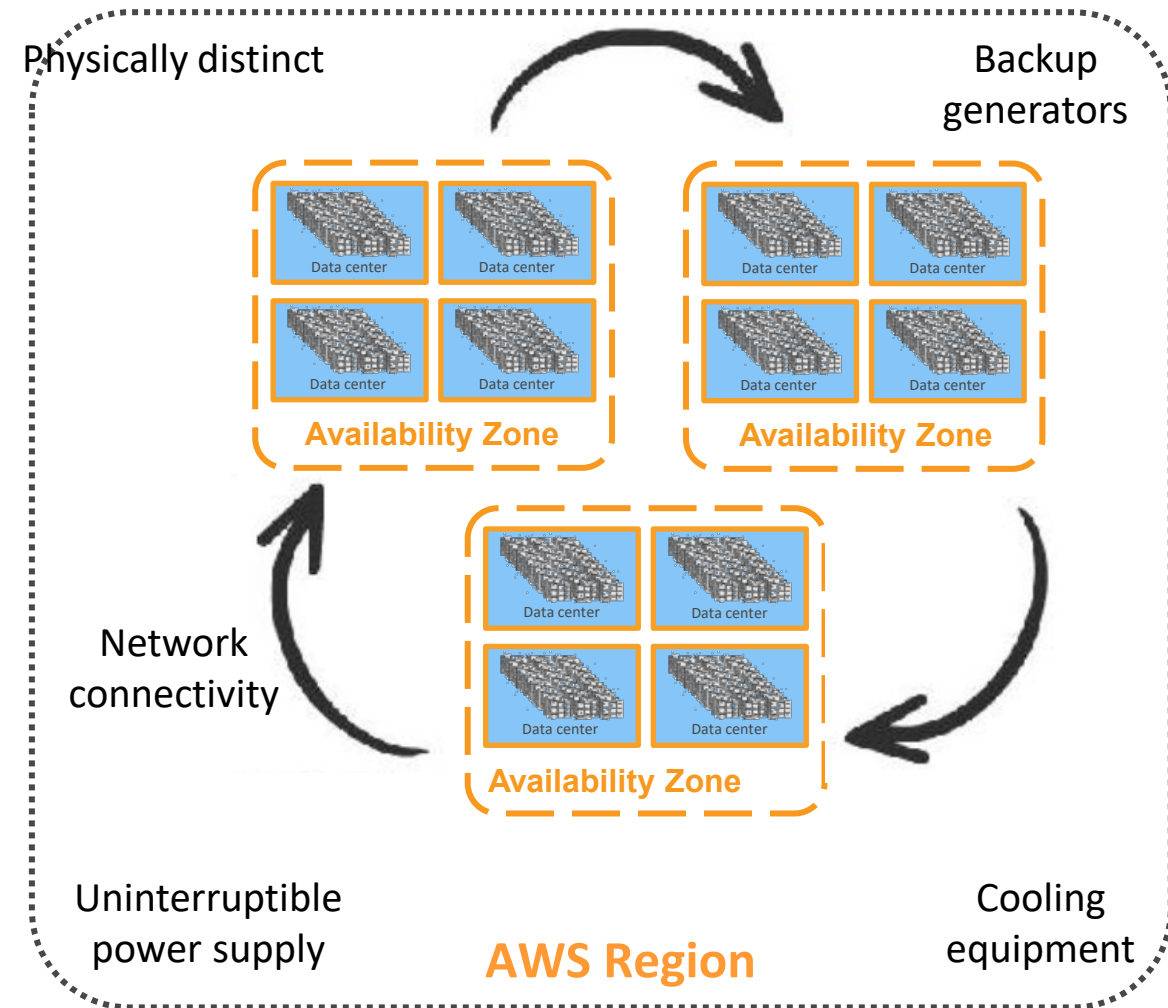
Points of Presence

- AWS provides a global network of **Points of Presence** locations
- Consists of **edge locations** and a much smaller number of **Regional edge caches**
- Used with Amazon CloudFront
 - A global Content Delivery Network (CDN), that delivers content to end users with **reduced latency**
- Regional edge caches used for content with infrequent access.



AWS infrastructure features

- Elasticity and scalability
 - Elastic infrastructure; dynamic adaption of capacity
 - Scalable infrastructure; adapts to accommodate growth
- Fault-tolerance
 - Continues operating properly in the presence of a failure
 - Built-in redundancy of components
- High availability
 - High level of operational performance
 - Minimized downtime
 - No human intervention



Key takeaways



- The **AWS Global Infrastructure** consists of **Regions** and **Availability Zones**.
- Your choice of a **Region** is typically based on **compliance requirements** or to **reduce latency**.
- Each **Availability Zone** is physically separate from other Availability Zones and has redundant power, networking, and connectivity.
- **Edge locations**, and **Regional edge caches** improve performance by **caching** content closer to users.



Section 2: AWS services and service category overview

Module 3: AWS Global Infrastructure Overview

AWS foundational services

Applications



Virtual desktops



Collaboration and sharing

Platform Services

Databases

Relational

NoSQL

Caching

Analytics

computing

Real-time

Data
warehouse

Data
workflows

Application services

Queuing

Orchestration

App Streaming

Transcoding

Email

Search

Deployment and management

Containers

DevOps tools

Resource templates

Usage tracking

Monitoring and logs

Mobile Services

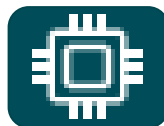
Identity

Sync

Mobile
Analytics

Notifications

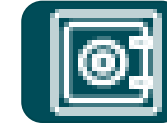
Foundation Services



Compute (virtual, automatic scaling, and load balancing)



Networking



Storage (object, block, and archive)

Infrastructure

Regions

Availability Zones



Edge locations

AWS categories of services



Analytics



Application
Integration



AR and VR



Blockchain



Business
Applications



Compute



Cost
Management



Customer
Engagement



Database



Developer Tools



End User
Computing



Game Tech



Internet
of Things



Machine
Learning



Management and
Governance



Media Services



Migration and
Transfer



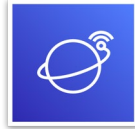
Mobile



Networking and
Content Delivery



Robotics



Satellite



Security, Identity, and
Compliance



Storage

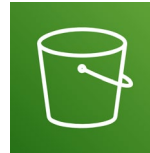
Storage service category



Photo from <https://www.pexels.com/photo/black-and-grey-device-159282/>



AWS storage services



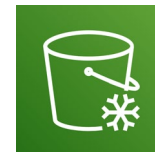
Amazon Simple Storage Service (Amazon S3)



Amazon Elastic Block Store (Amazon EBS)



Amazon Elastic File System (Amazon EFS)



Amazon Simple Storage Service Glacier

Compute service category

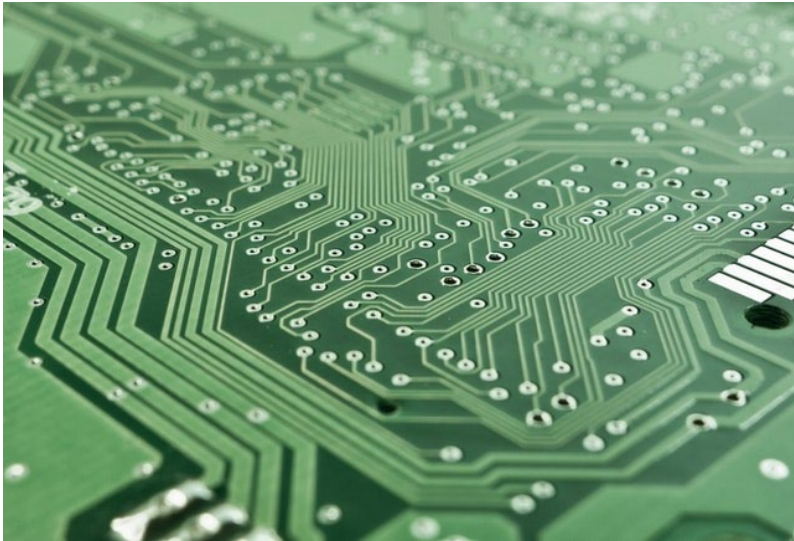
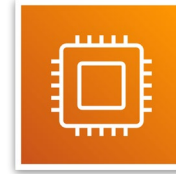
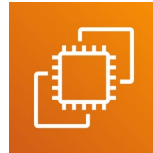


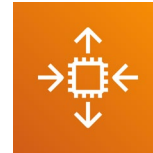
Photo from <https://www.pexels.com/photo/technology-computer-lines-board-50711/>



AWS Compute services



Amazon EC2



Amazon EC2
Auto Scaling



Amazon Elastic
Container Service
(Amazon ECS)



Amazon EC2
Container Registry



AWS Elastic
Beanstalk



AWS Lambda



Amazon Elastic
Kubernetes Service
(Amazon EKS)

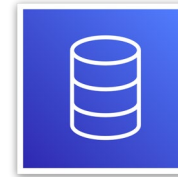


AWS Fargate

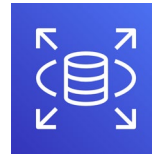
Database service category



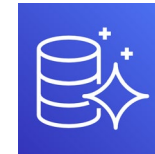
Photo from <https://aws.amazon.com/compliance/data-center/data-centers/>



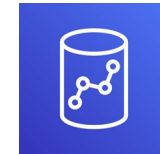
AWS Database services



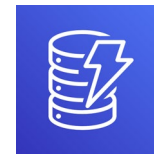
Amazon Relational
Database Service



Amazon Aurora



Amazon Redshift

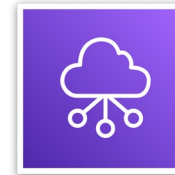


Amazon
DynamoDB

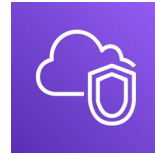
Networking and content delivery service category



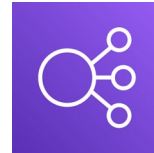
Photo by Umberto on Unsplash



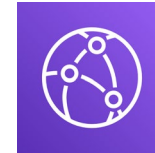
**AWS networking
and content delivery services**



Amazon VPC



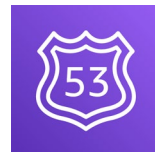
Elastic Load
Balancing



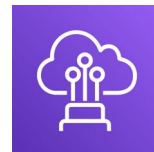
Amazon
CloudFront



AWS Transit
Gateway



Amazon
Route 53



AWS Direct
Connect



AWS VPN

Security, identity, and compliance service category



Photo by Paweł Czerwiński on Unsplash



**AWS security, identity,
and compliance services**



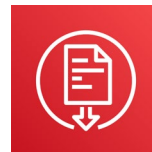
AWS Identity and Access
Management (IAM)



AWS
Organizations



Amazon Cognito



AWS Artifact



AWS Key
Management
Service



AWS Shield

AWS cost management service category



Photo by Alexander Mills on Unsplash



AWS cost management
services



AWS Cost and
Usage Report



AWS Budgets



AWS Cost
Explorer

Management and governance service category

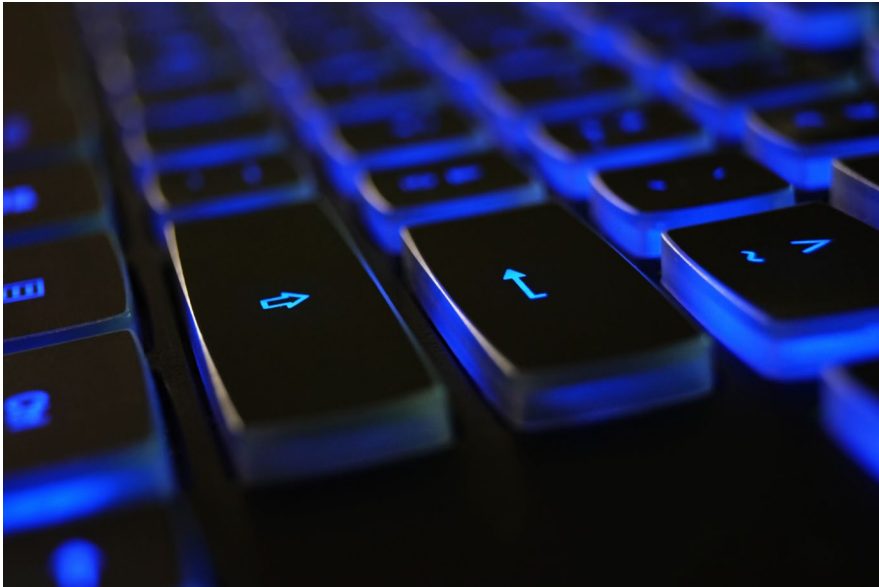
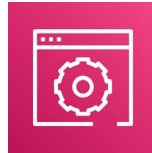


Photo by Marta Branco from Pexels



AWS management and governance services



AWS Management Console



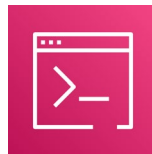
AWS Config



Amazon CloudWatch



AWS Auto Scaling



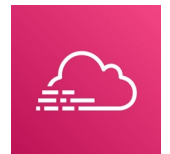
AWS Command Line Interface



AWS Trusted Advisor



AWS Well-Architected Tool



AWS CloudTrail



Module wrap-up

Module 3: AWS Global Infrastructure Overview



Module summary

In summary, in this module you learned how to:

- Identify the difference between AWS Regions, Availability Zones, and edge locations
- Identify AWS service and service categories

Complete the knowledge check





Sample exam question



Which component of AWS global infrastructure does Amazon CloudFront use to ensure low-latency delivery?

Choice	Response
A	AWS Regions
B	AWS edge locations
C	AWS Availability Zones
D	Amazon Virtual Private Cloud (Amazon VPC)



Additional resources

- AWS Global Infrastructure: <https://aws.amazon.com/about-aws/global-infrastructure/>
- AWS Regional Services List: <https://aws.amazon.com/about-aws/global-infrastructure/regional-product-services/>
- AWS Cloud Products: <https://aws.amazon.com/products/>



Module 4: AWS Cloud Security

AWS Academy Cloud Foundations



Module overview

Topics

- AWS shared responsibility model
- AWS Identity and Access Management (IAM)
- Securing a new AWS account
- Securing accounts
- Securing data on AWS
- Working to ensure compliance

Activities

- AWS shared responsibility model activity

Demo

- Recorded demonstration of IAM

Lab

- Introduction to AWS IAM



Knowledge check



Module objectives

After completing this module, you should be able to:

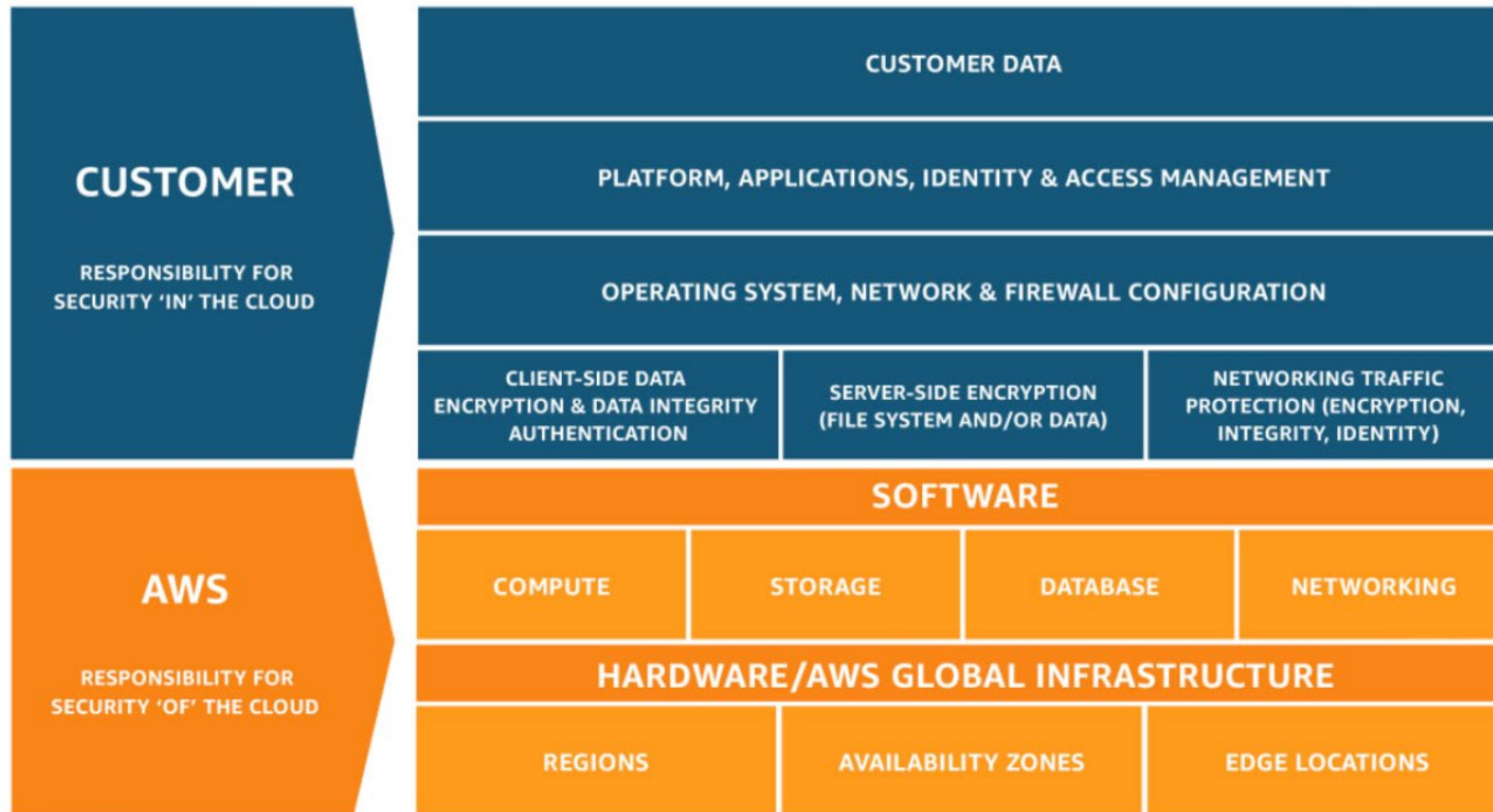
- Recognize the shared responsibility model
- Identify the responsibility of the customer and AWS
- Recognize IAM users, groups, and roles
- Describe different types of security credentials in IAM
- Identify the steps to securing a new AWS account
- Explore IAM users and groups
- Recognize how to secure AWS data
- Recognize AWS compliance programs



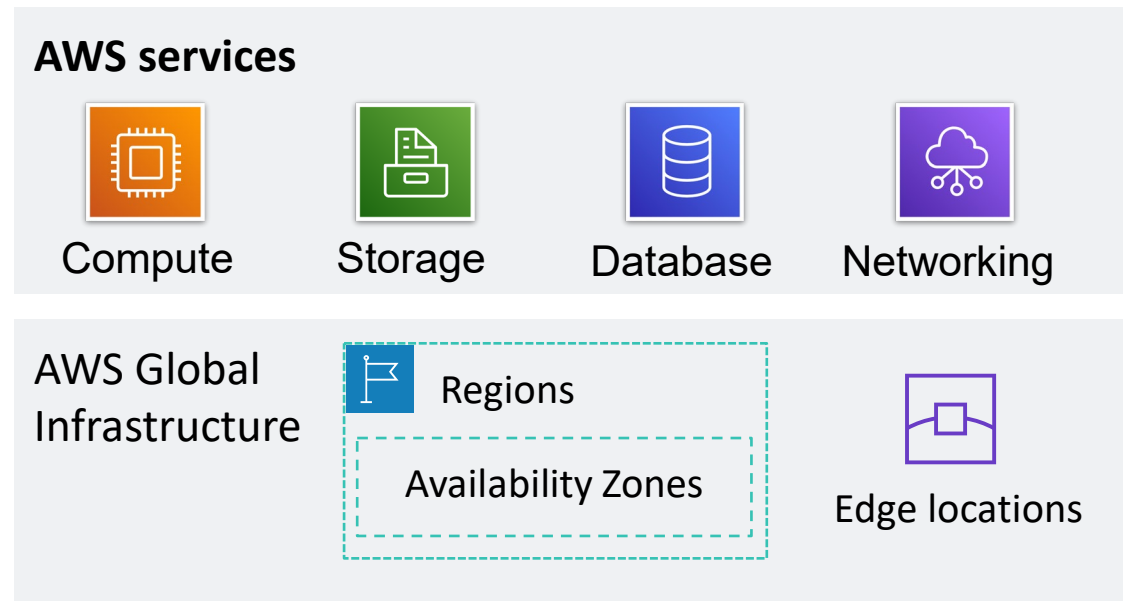
Section 1: AWS shared responsibility model

Module 4: AWS Cloud Security

AWS shared responsibility model



AWS responsibility: Security *of* the cloud

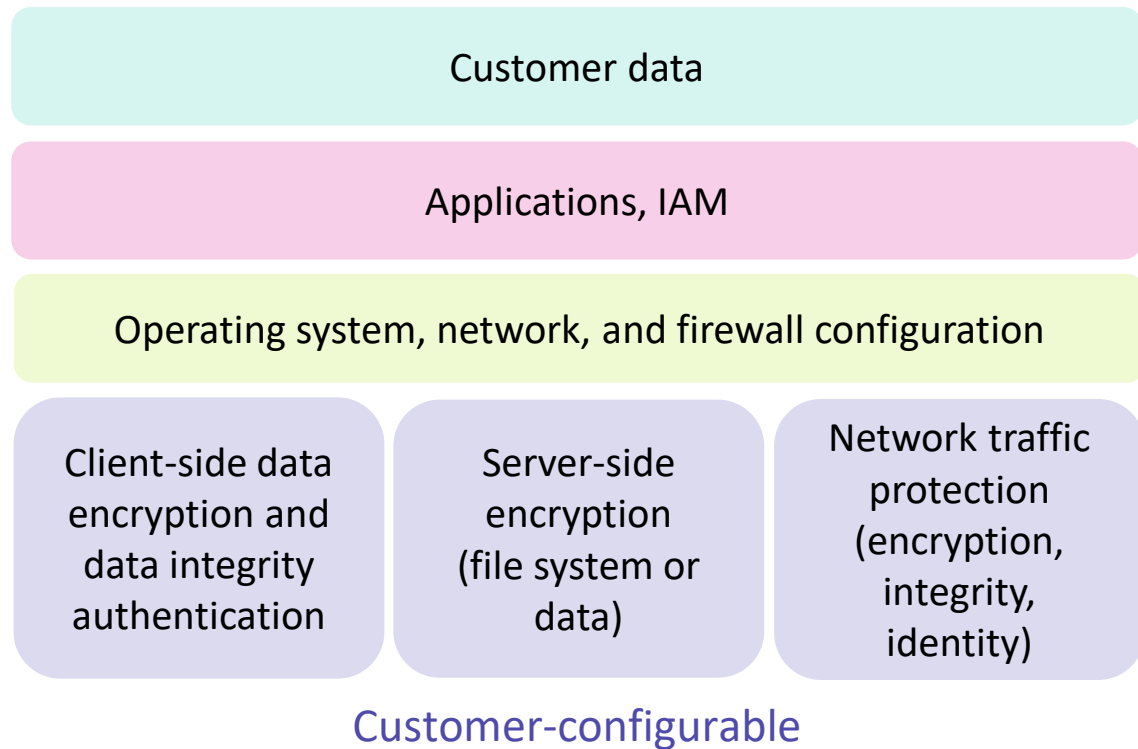


AWS responsibilities:

- Physical security of data centers
 - Controlled, need-based access
- Hardware and software infrastructure
 - Storage decommissioning, host operating system (OS) access logging, and auditing
- Network infrastructure
 - Intrusion detection
- Virtualization infrastructure
 - Instance isolation



Customer responsibility: Security *in* the cloud

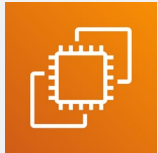


Customer responsibilities:

- Amazon Elastic Compute Cloud (Amazon EC2) instance **operating system**
 - Including patching, maintenance
- **Applications**
 - Passwords, role-based access, etc.
- **Security group** configuration
- OS or host-based **firewalls**
 - Including intrusion detection or prevention systems
- **Network** configurations
- Account management
 - Login and permission settings for each user

Service characteristics and security responsibility (1 of 2)

Example services managed by the customer



Amazon
EC2



Amazon Elastic
Block Store
(Amazon EBS)

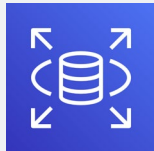


Amazon
Virtual Private Cloud
(Amazon VPC)

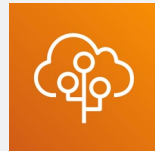
Example services managed by AWS



AWS
Lambda



Amazon
Relational Database
Service (Amazon RDS)



AWS Elastic
Beanstalk

Infrastructure as a service (IaaS)

- Customer has more flexibility over configuring networking and storage settings
- Customer is responsible for managing more aspects of the security
- Customer configures the access controls

Platform as a service (PaaS)

- Customer does not need to manage the underlying infrastructure
- AWS handles the operating system, database patching, firewall configuration, and disaster recovery
- Customer can focus on managing code or data

Service characteristics and security responsibility (2 of 2)

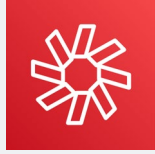
SaaS examples



AWS Trusted
Advisor



AWS Shield



Amazon Chime

Software as a service (SaaS)

- Software is centrally hosted
- Licensed on a subscription model or pay-as-you-go basis.
- Services are typically accessed via web browser, mobile app, or application programming interface (API)
- Customers do not need to manage the infrastructure that supports the service

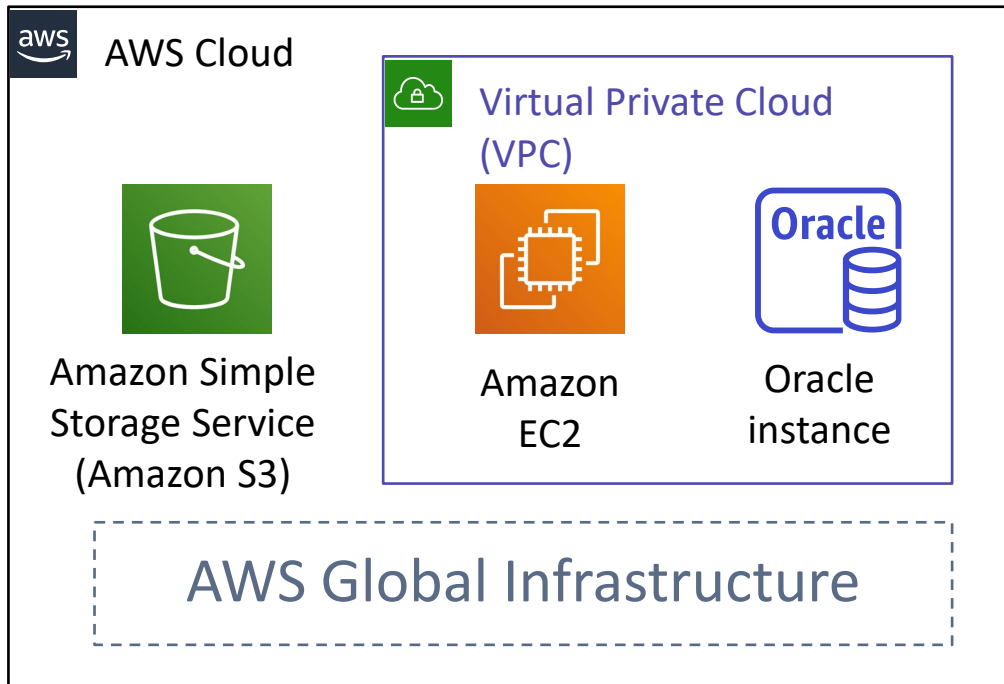
Activity: AWS shared responsibility model



Photo by Pixabay from Pexels.

Activity: Scenario 1 of 2

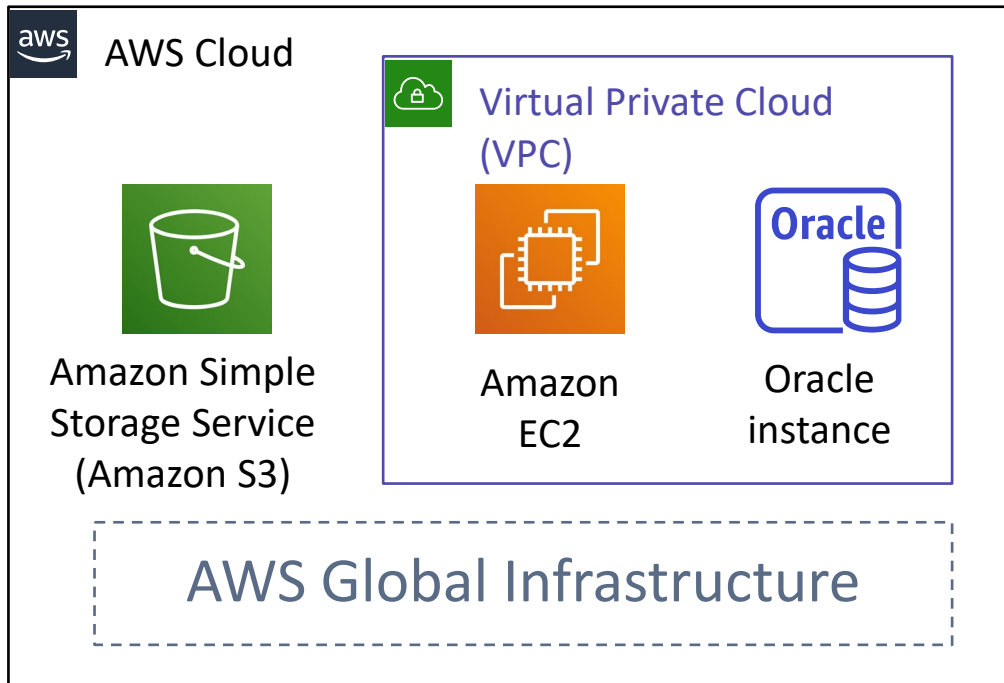
Consider this deployment. Who is responsible – AWS or the customer?



1. Upgrades and patches to the operating system on the EC2 instance?
2. Physical security of the data center?
3. Virtualization infrastructure?
4. EC2 security group settings?
5. Configuration of applications that run on the EC2 instance?
6. Oracle upgrades or patches If the Oracle instance runs as an Amazon RDS instance?
7. Oracle upgrades or patches If Oracle runs on an EC2 instance?
8. S3 bucket access configuration?

Activity: Scenario 1 of 2 Answers

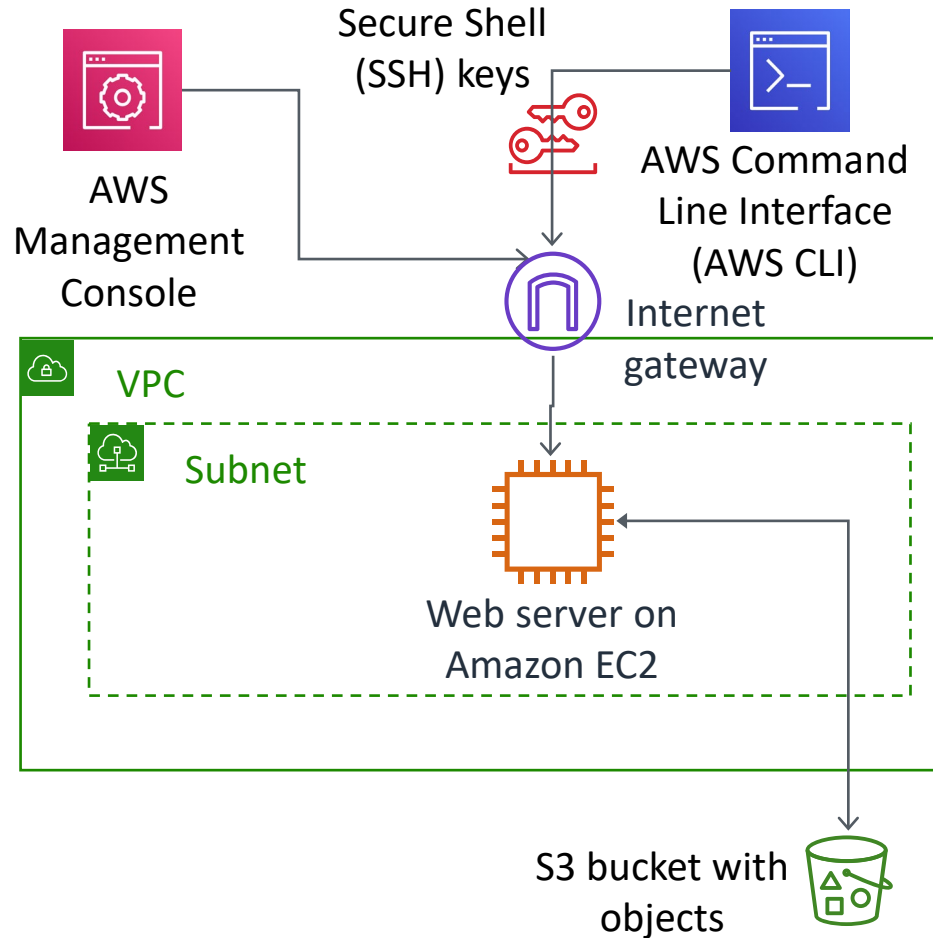
Consider this deployment. Who is responsible – AWS or the customer?



1. Upgrades and patches to the operating system on the EC2 instance?
 - **ANSWER:** The customer
2. Physical security of the data center?
 - **ANSWER:** AWS
3. Virtualization infrastructure?
 - **ANSWER:** AWS
4. EC2 security group settings?
 - **ANSWER:** The customer
5. Configuration of applications that run on the EC2 instance?
 - **ANSWER:** The customer
6. Oracle upgrades or patches If the Oracle instance runs as an Amazon RDS instance?
 - **ANSWER:** AWS
7. Oracle upgrades or patches If Oracle runs on an EC2 instance?
 - **ANSWER:** The customer
8. S3 bucket access configuration?
 - **ANSWER:** The customer

Activity: Scenario 2 of 2

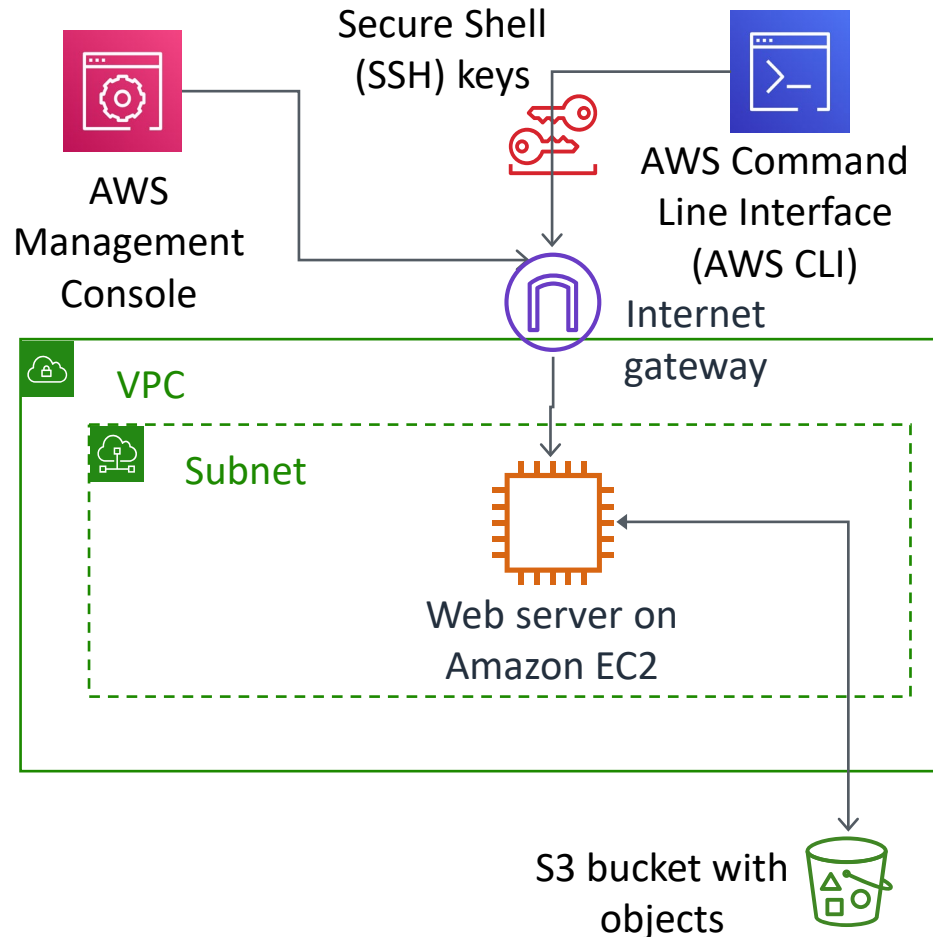
Consider this deployment. Who is responsible – AWS or the customer?



1. Ensuring that the AWS Management Console is not hacked?
2. Configuring the subnet?
3. Configuring the VPC?
4. Protecting against network outages in AWS Regions?
5. Securing the SSH keys
6. Ensuring network isolation between AWS customers' data?
7. Ensuring low-latency network connection between the web server and the S3 bucket?
8. Enforcing multi-factor authentication for all user logins?

Activity: Scenario 2 of 2 Answers

Consider this deployment. Who is responsible – AWS or the customer?



1. Ensuring that the AWS Management Console is not hacked?
 - **ANSWER:** AWS
2. Configuring the subnet?
 - **ANSWER:** The customer
3. Configuring the VPC?
 - **ANSWER:** The customer
4. Protecting against network outages in AWS Regions?
 - **ANSWER:** AWS
5. Securing the SSH keys
 - **ANSWER:** The customer
6. Ensuring network isolation between AWS customers' data?
 - **ANSWER:** AWS
7. Ensuring low-latency network connection between the web server and the S3 bucket?
 - **ANSWER:** AWS
8. Enforcing multi-factor authentication for all user logins?
 - **ANSWER:** The customer

Section 1 key takeaways



- AWS and the customer share security responsibilities:
 - AWS is responsible for security **of** the cloud
 - Customer is responsible for security **in** the cloud
- **AWS is responsible for protecting the infrastructure**—including hardware, software, networking, and facilities—that run AWS Cloud services
- For services that are categorized as infrastructure as a service (IaaS), the **customer is responsible for performing necessary security configuration and management tasks**
 - For example, guest OS updates and security patches, firewall, security group configurations



Section 2: AWS Identity and Access Management (IAM)

Module 4: AWS Cloud Security



AWS Identity and Access Management (IAM)

- Use **IAM** to manage access to **AWS resources** –
 - A resource is an entity in an AWS account that you can work with
 - Example resources; An Amazon EC2 instance or an Amazon S3 bucket
- *Example* – Control who can terminate Amazon EC2 instances
- Define fine-grained access rights –
 - **Who** can access the resource
 - **Which** resources can be accessed and what can the user do to the resource
 - **How** resources can be accessed
- IAM is a no-cost AWS account feature



AWS Identity and Access
Management
(IAM)

IAM: Essential components



IAM user

A **person** or **application** that can authenticate with an AWS account.



IAM group

A **collection of IAM users** that are granted identical authorization.



IAM policy

The document that defines **which resources can be accessed** and the **level of access** to each resource.



IAM role

Useful mechanism to grant a set of permissions for making AWS service requests.



Authenticate as an IAM user to gain access

When you define an **IAM user**, you select what *types of access* the user is permitted to use.

Programmatic access

- Authenticate using:
 - Access key ID
 - Secret access key
- Provides AWS CLI and AWS SDK access



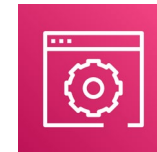
AWS CLI



AWS Tools
and SDKs

AWS Management Console access

- Authenticate using:
 - 12-digit Account ID *or* alias
 - IAM user name
 - IAM password
- If enabled, **multi-factor authentication (MFA)** prompts for an authentication code.



AWS Management
Console

IAM MFA

- MFA provides increased security.
- In addition to **username** and **password**, MFA requires a unique **authentication code** to access AWS services.

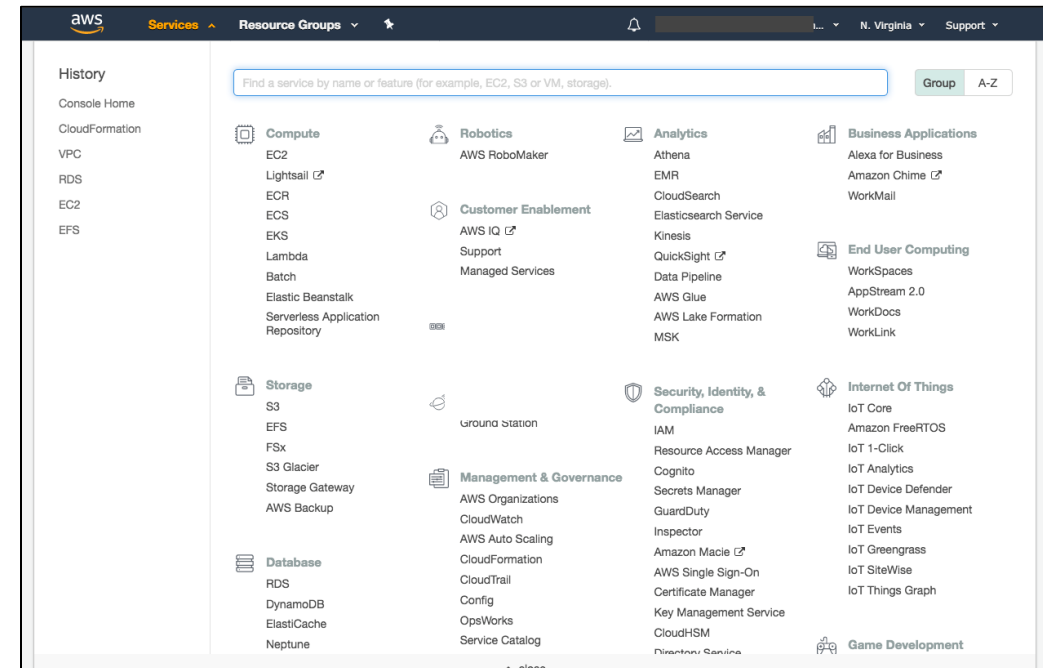
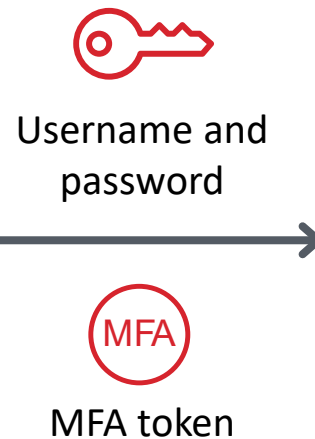
Account:

User Name:

Password:

MFA users, enter your code on the next screen.

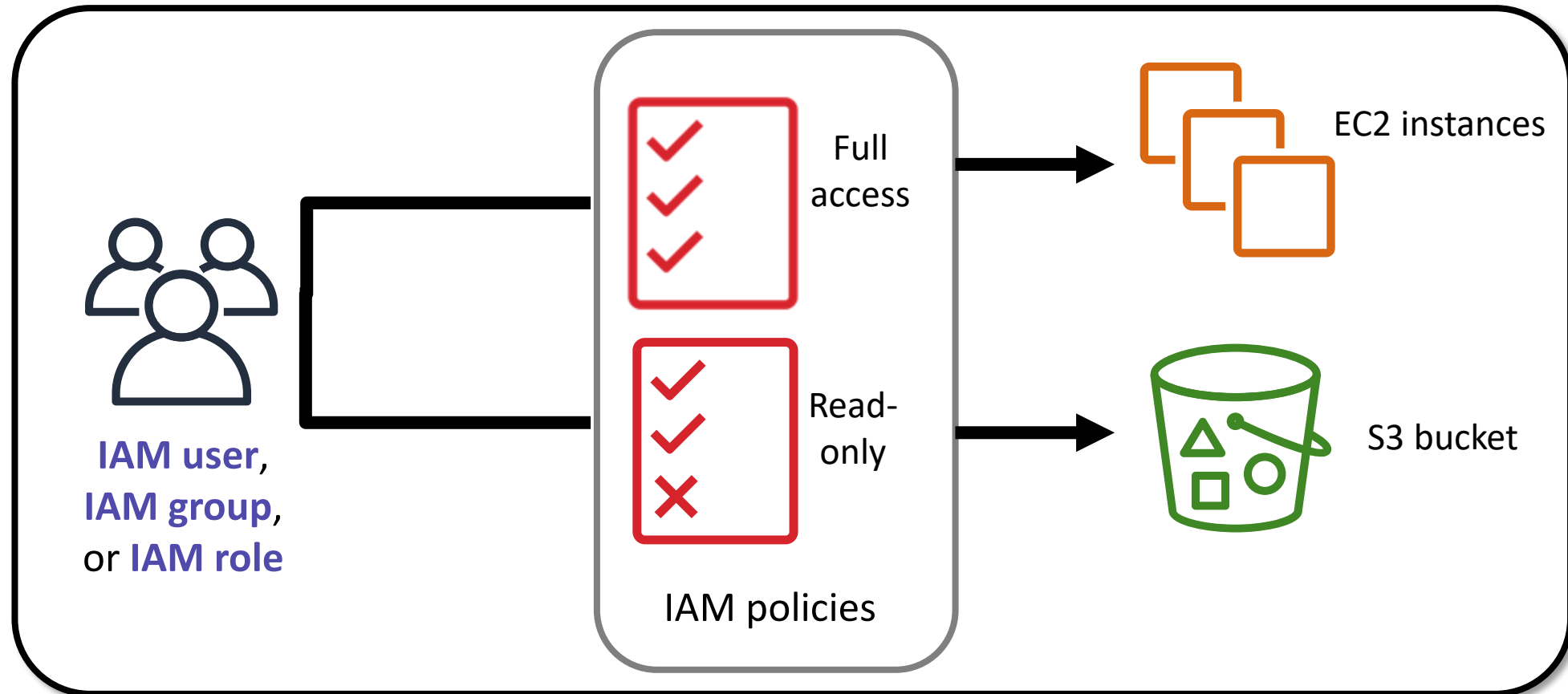
[Sign In](#)



AWS Management Console

Authorization: What actions are permitted

After the user or application is connected to the AWS account, what are they allowed to do?





IAM: Authorization

- Assign permissions by creating an IAM policy.
- Permissions determine **which resources and operations** are allowed:
 - All permissions are implicitly denied by default.
 - If something is explicitly denied, it is never allowed.

Best practice: Follow the **principle of least privilege**.

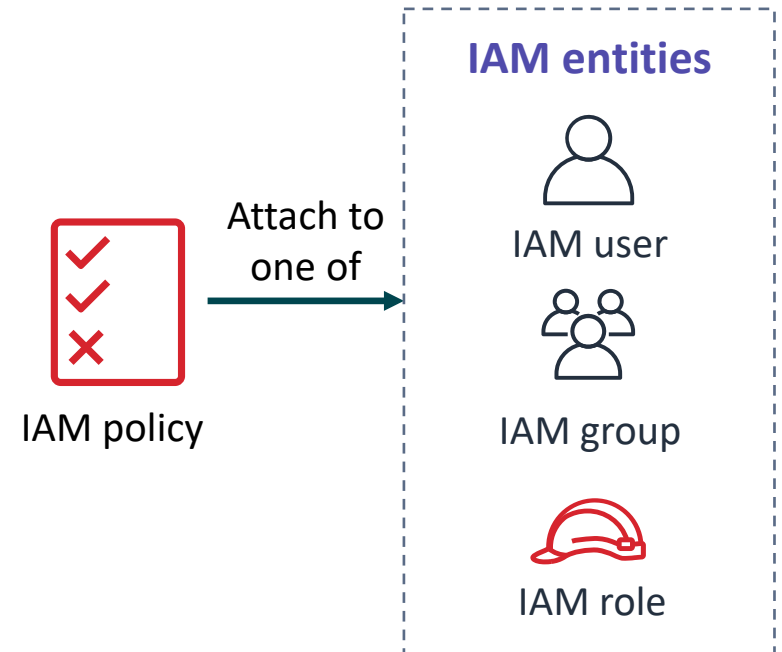


**IAM
permissions**

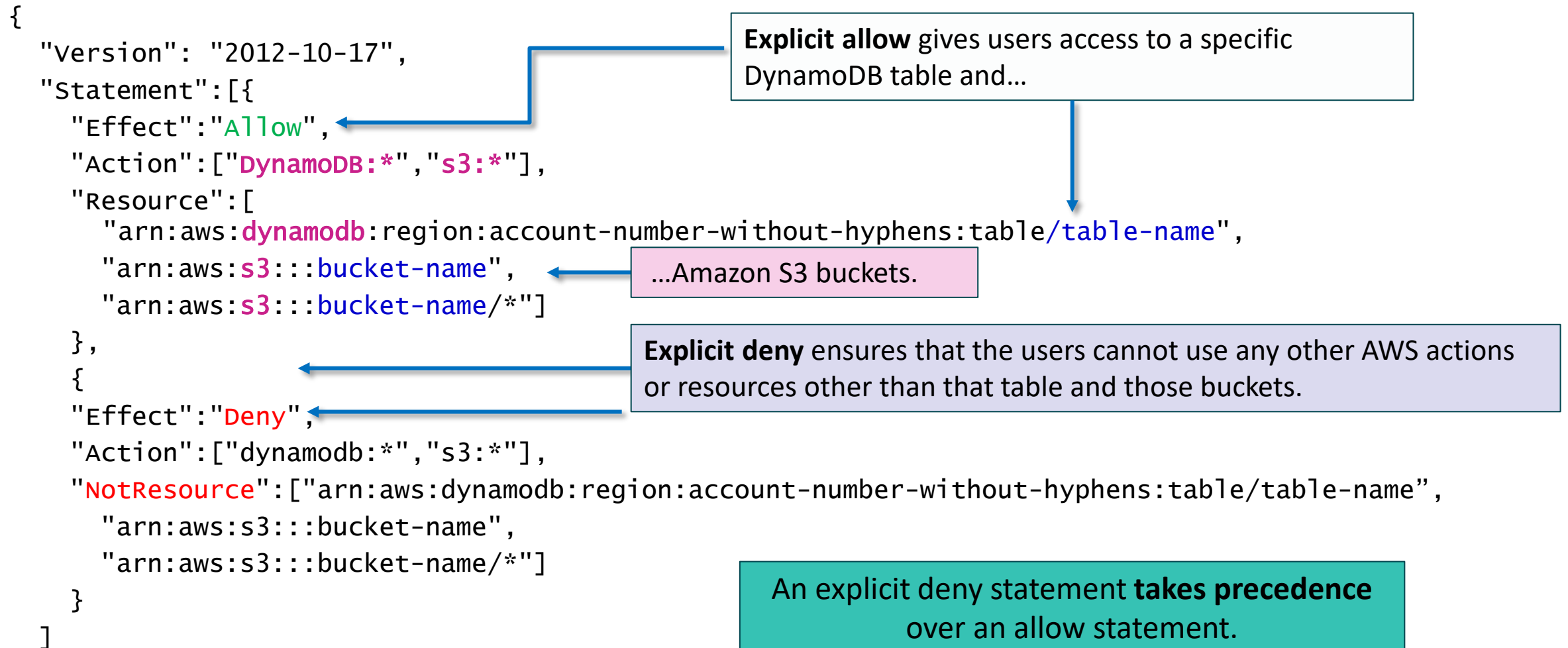
Note: The scope of IAM service configurations is **global**. Settings apply across all AWS Regions.

IAM policies

- An IAM policy is a document that defines permissions
 - Enables fine-grained access control
- Two types of policies – *identity-based* and *resource-based*
- **Identity-based** policies –
 - Attach a policy to any IAM entity
 - An IAM user, an IAM group, or an IAM role
 - Policies specify:
 - Actions that **may** be performed by the entity
 - Actions that **may not** be performed by the entity
 - A single *policy* can be attached to multiple *entities*
 - A single *entity* can have multiple *policies* attached to it
- **Resource-based** policies
 - Attached to a resource (such as an S3 bucket)

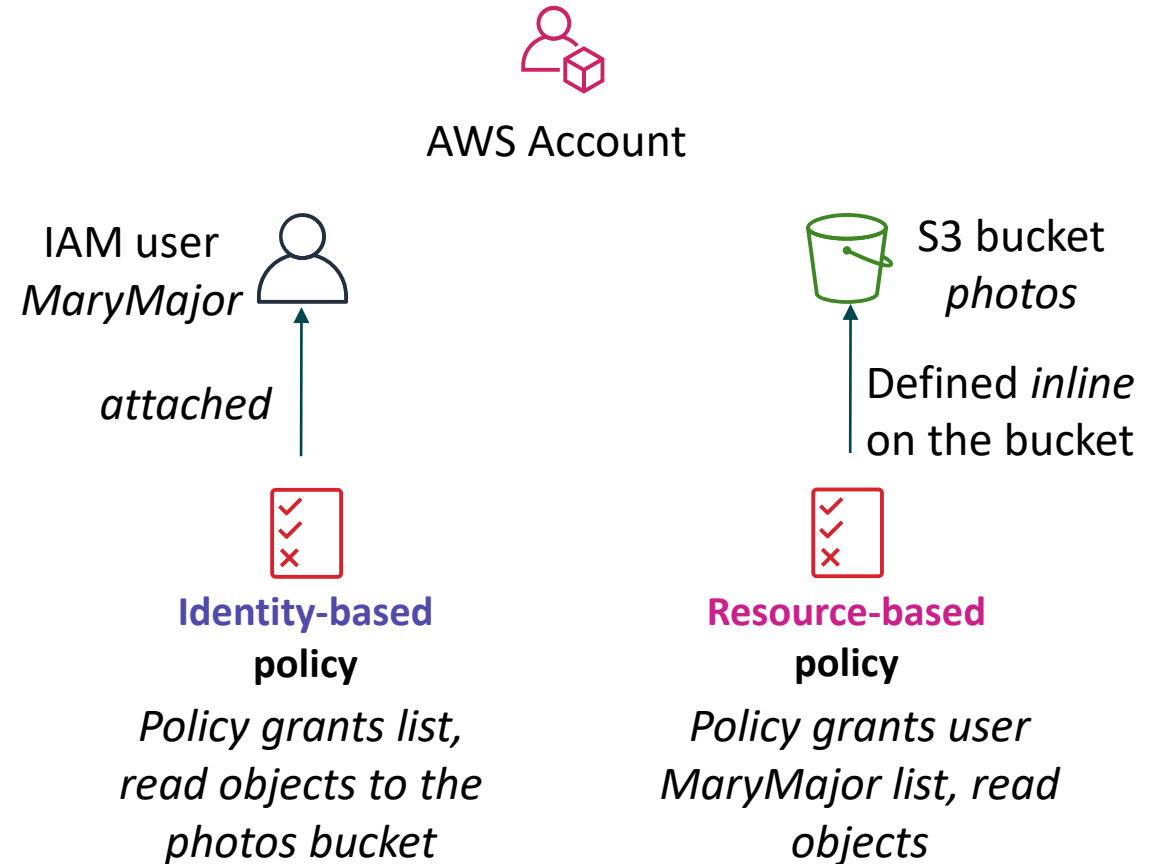


IAM policy example



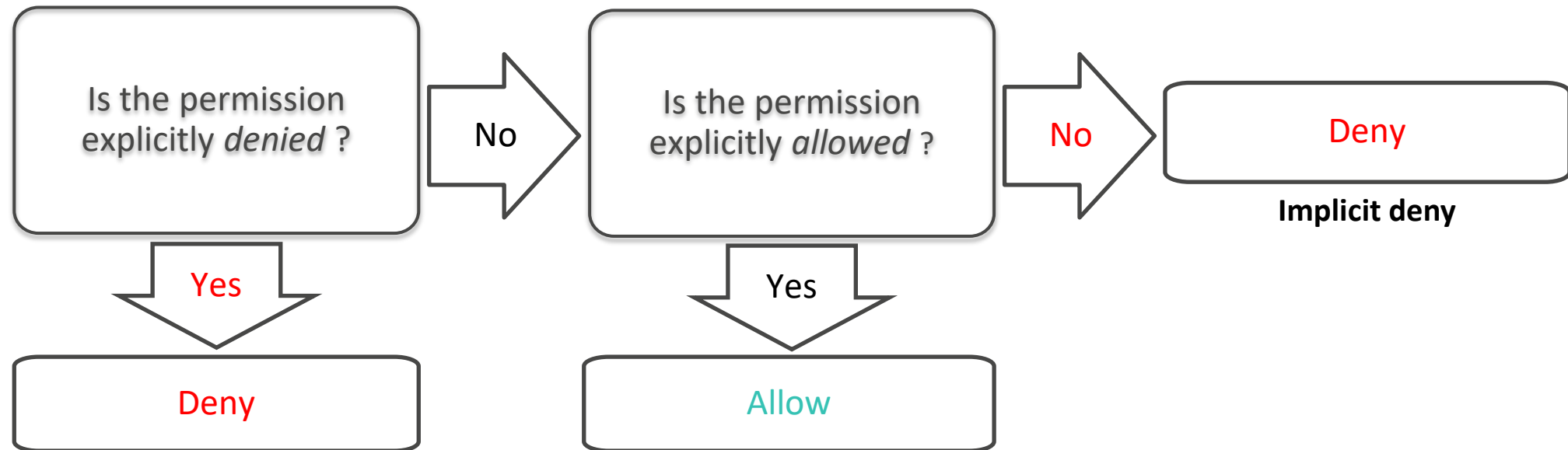
Resource-based policies

- *Identity-based policies* are attached to a user, group, or role
- **Resource-based policies** are attached to a resource (*not* to a user, group or role)
- Characteristics of resource-based policies –
 - Specifies who has access to the resource and what actions they can perform on it
 - The policies are *inline* only, not managed
- Resource-based policies are supported only by some AWS services



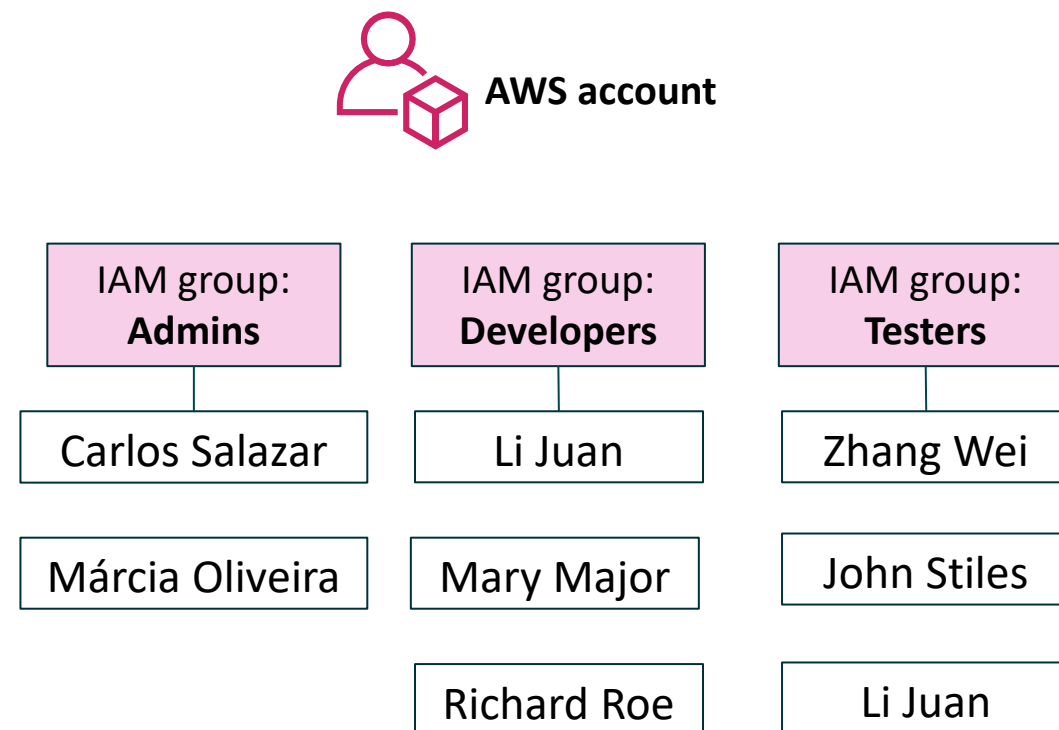
IAM permissions

How IAM determines permissions:



IAM groups

- An **IAM group** is a collection of IAM users
- A group is used to grant the same permissions to multiple users
 - Permissions granted by attaching IAM *policy* or policies to the group
- A user can belong to multiple groups
- There is no default group
- Groups cannot be nested





IAM roles

- An **IAM role** is an IAM identity with specific permissions
- Similar to an IAM user
 - Attach permissions policies to it
- Different from an IAM user
 - Not uniquely associated with one person
 - Intended to be *assumable* by a **person**, **application**, or **service**
- Role provides *temporary* security credentials
- Examples of how IAM roles are used to **delegate** access –
 - Used by an IAM user in the same AWS account as the role
 - Used by an AWS service—such as Amazon EC2—in the same account as the role
 - Used by an IAM user in a different AWS account than the role



IAM role

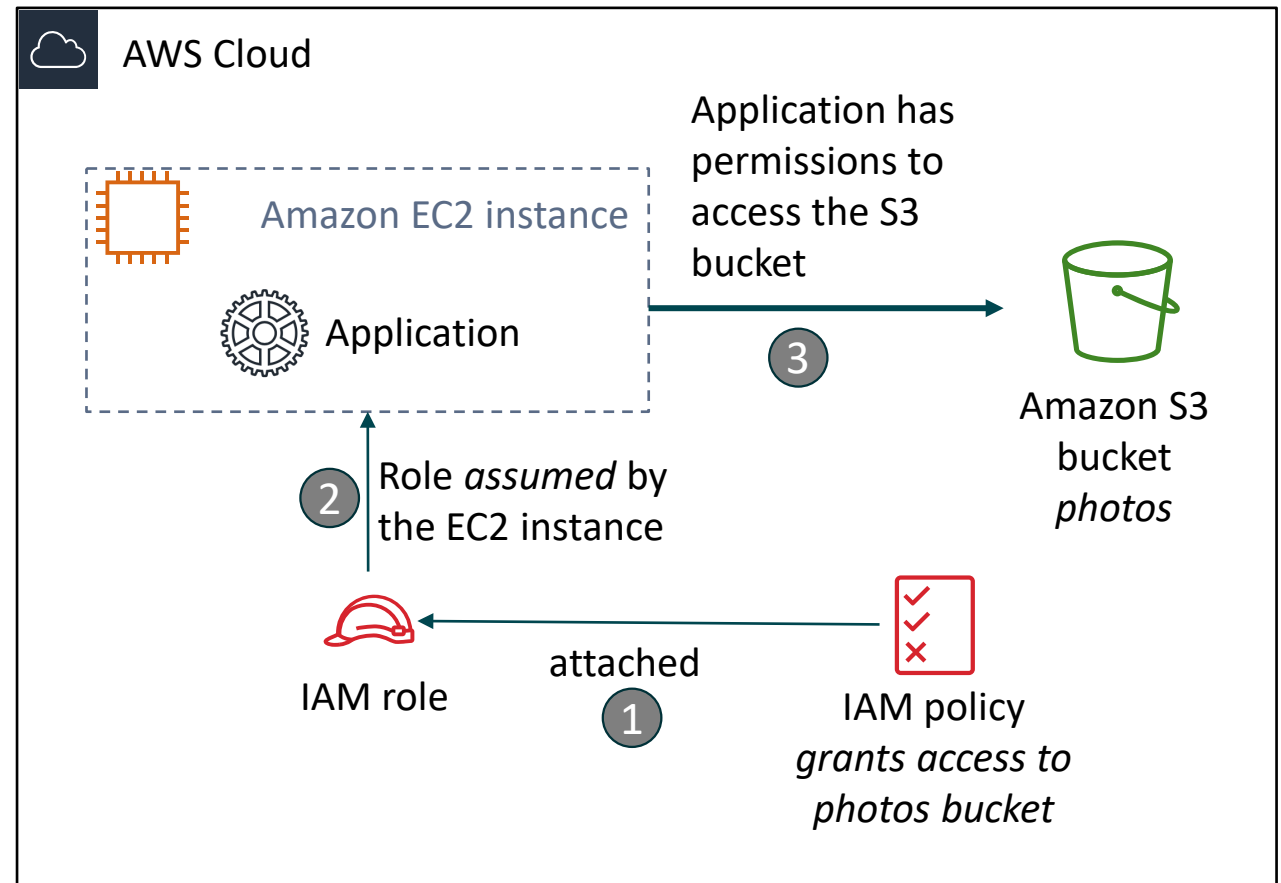
Example use of an IAM role

Scenario:

- An application that runs on an EC2 instance needs access to an S3 bucket

Solution:

- Define an IAM policy that grants access to the S3 bucket.
- Attach the policy to a role
- Allow the EC2 instance to assume the role

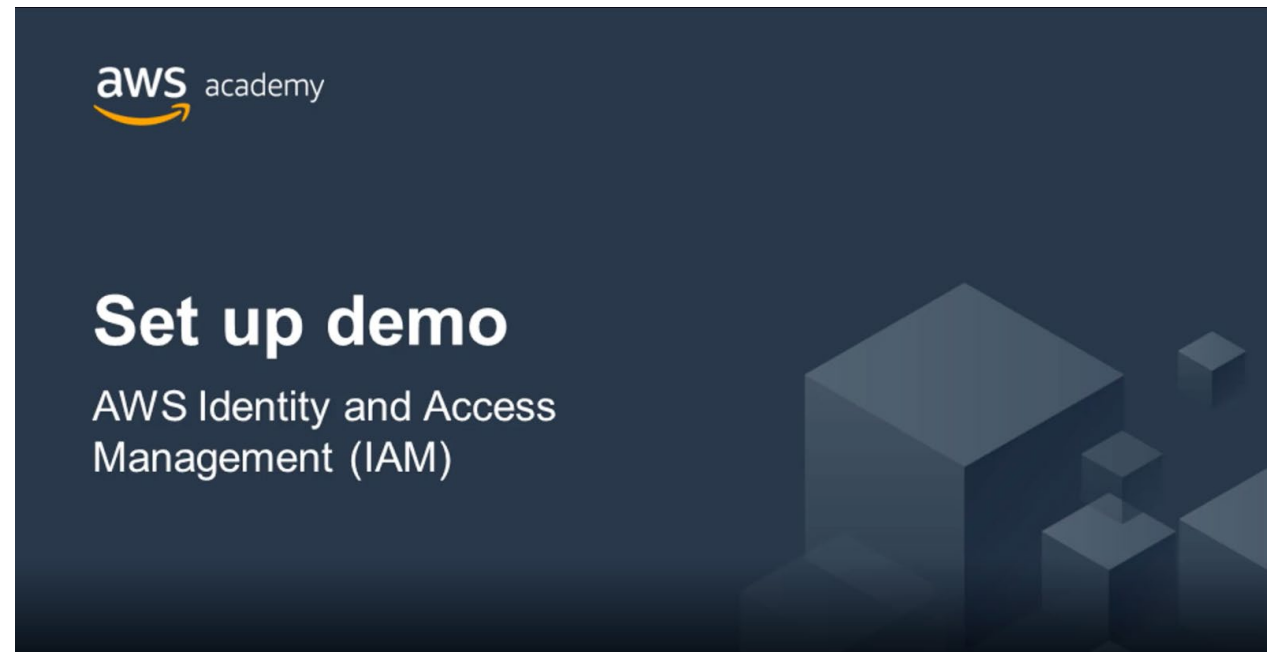


Section 2 key takeaways



- **IAM policies** are constructed with JavaScript Object Notation (JSON) and define permissions.
 - IAM policies can be attached to any **IAM entity**.
 - Entities are IAM users, IAM groups, and IAM roles.
- An **IAM user** provides a way for a person, application, or service to authenticate to AWS.
- An **IAM group** is a simple way to attach the same policies to multiple users.
- An **IAM role** can have permissions policies attached to it and can be used to delegate temporary access to users or applications.

Recorded demo: IAM

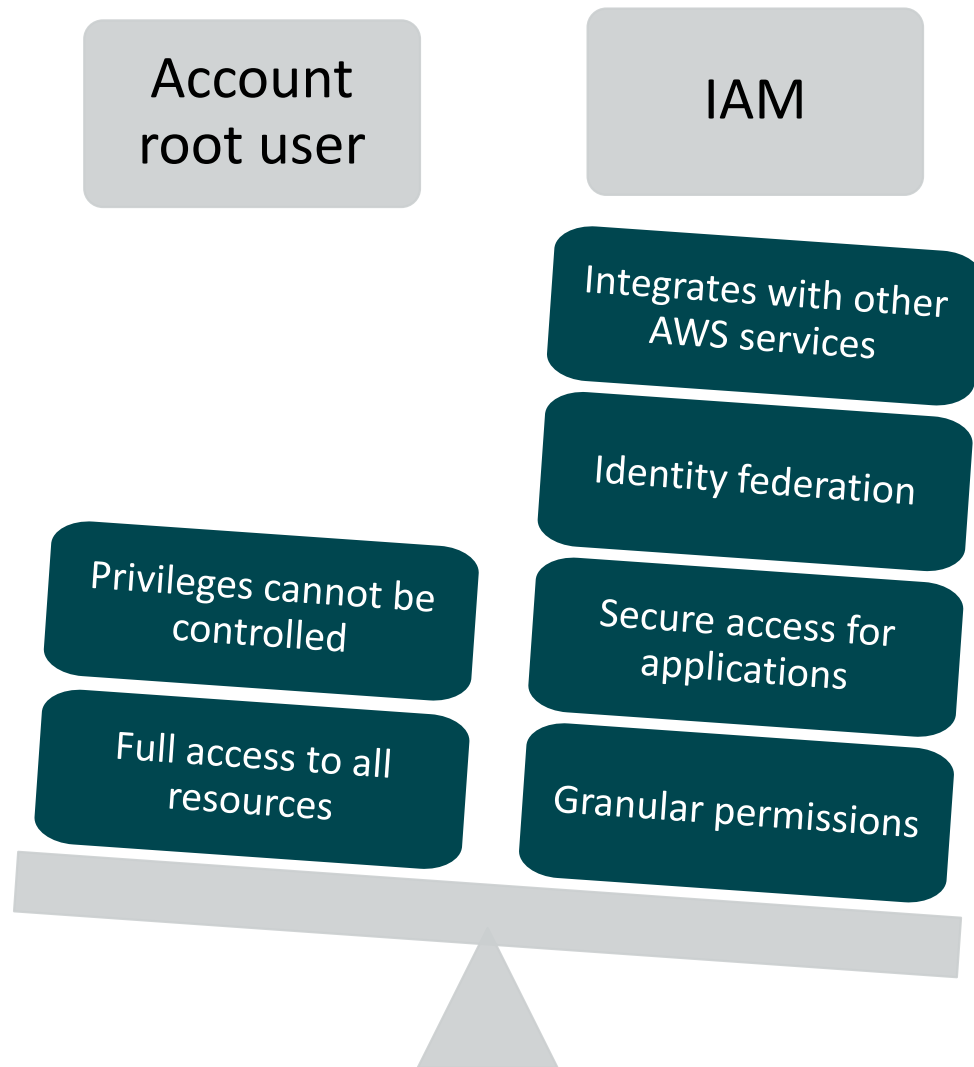




Section 3: Securing a new AWS account

Module 4: AWS Cloud Security

AWS account root user access versus IAM access



- **Best practice:** Do not use the AWS account root user except when necessary.
 - Access to the **account root user** requires logging in with the *email address* (and password) that you used to create the account.
- Example actions that can only be done with the account root user:
 - Update the account root user password
 - Change the AWS Support plan
 - Restore an IAM user's permissions
 - Change account settings (for example, contact information, allowed Regions)



Securing a new AWS account: Account root user

Step 1: Stop using the account root user as soon as possible.

- The account root user has unrestricted access to all your resources.
- To stop using the account root user:
 1. While you are logged in as the account root user, **create an IAM user** for yourself. Save the access keys if needed.
 2. Create an IAM group, give it full administrator permissions, and add the IAM user to the group.
 3. Disable and **remove your account root user access keys**, if they exist.
 4. **Enable a password policy** for users.
 5. Sign in with your new IAM user credentials.
 6. Store your account root user credentials in a secure place.



Securing a new AWS account: MFA

Step 2: Enable multi-factor authentication (MFA).

- Require MFA for your [account root user](#) and for [all IAM users](#).
- You can also use MFA to control access to AWS service APIs.
- Options for retrieving the MFA token –
 - Virtual MFA-compliant applications:
 - Google Authenticator.
 - Authy Authenticator (Windows phone app).
 - U2F security key devices:
 - For example, YubiKey.
 - Hardware MFA options:
 - Key fob or display card offered by [Gemalto](#).



MFA token



Securing a new AWS account: AWS CloudTrail

Step 3: Use AWS CloudTrail.

- CloudTrail tracks user activity on your account.
 - Logs all API requests to resources in all supported services your account.
- Basic AWS CloudTrail event history is enabled by default and is free.
 - It contains all management event data on latest 90 days of account activity.
- To access CloudTrail –
 1. Log in to the **AWS Management Console** and choose the **CloudTrail** service.
 2. Click **Event history** to view, filter, and search the last 90 days of events.
- To enable logs beyond 90 days and enable specified event alerting, create a trail.
 1. From the CloudTrail Console trails page, click **Create trail**.
 2. Give it a name, apply it to all Regions, and create a new Amazon S3 bucket for log storage.
 3. Configure access restrictions on the S3 bucket (for example, only admin users should have access).



Securing a new AWS account: Billing reports

Step 4: Enable a billing report, such as the AWS Cost and Usage Report.

- Billing reports provide information about your use of AWS resources and estimated costs for that use.
- AWS delivers the reports to an Amazon S3 bucket that you specify.
 - Report is updated at least once per day.
- The **AWS Cost and Usage Report** tracks your AWS usage and provides estimated charges associated with your AWS account, either by the hour or by the day.

Section 3 key takeaways



Best practices to secure an AWS account:

- **Secure** logins with multi-factor authentication (MFA).
- **Delete** account root user **access keys**.
- **Create** individual **IAM users** and grant permissions according to the principle of least privilege.
- **Use groups** to assign permissions to IAM users.
- **Configure** a **strong password policy**.
- **Delegate** using **roles** instead of sharing credentials.
- **Monitor** account activity by using AWS CloudTrail.

Lab 1:

Introduction to IAM





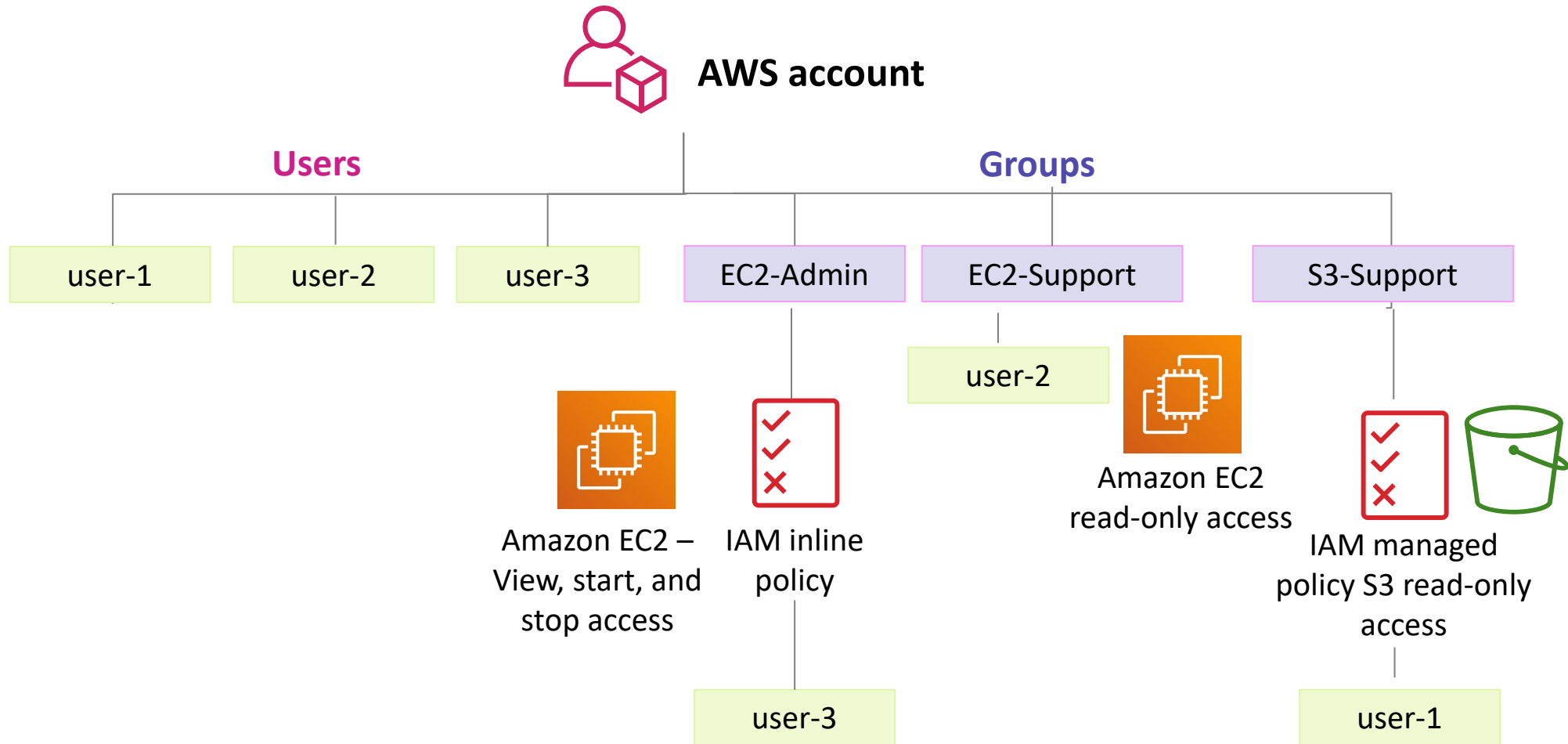
Lab 1: Tasks

- Task 1: Explore the Users and Groups.
- Task 2: Add Users to Groups.
- Task 3: Sign-In and Test Users.



AWS Identity and Access
Management (IAM)

Lab 1: Final product





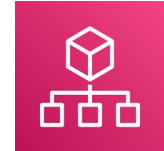
Section 4: Securing accounts

Module 4: AWS Cloud Security



AWS Organizations

- **AWS Organizations** enables you to consolidate multiple AWS accounts so that you centrally manage them.
- **Security features** of AWS Organizations:
 - **Group AWS accounts into organizational units (OUs)** and attach different access policies to each OU.
 - **Integration and support for IAM**
 - Permissions to a user are the intersection of what is allowed by AWS Organizations and what is granted by IAM in that account.
 - **Use service control policies** to establish control over the AWS services and API actions that each AWS account can access



AWS Organizations



AWS Organizations: Service control policies

- **Service control policies (SCPs)** offer centralized control over accounts.
 - Limit permissions that are available in an account that is part of an organization.
- Ensures that accounts comply with access control guidelines.
- SCPs are *similar* to IAM permissions policies –
 - They use similar syntax.
 - However, an SCP never grants permissions.
 - Instead, SCPs **specify the maximum permissions** for an organization.



AWS Key Management Service (AWS KMS)

AWS Key Management Service (AWS KMS) features:

- Enables you to **create and manage encryption keys**
- Enables you to control the use of encryption across AWS services and in your applications.
- Integrates with AWS CloudTrail to log all key usage.
- Uses hardware security modules (HSMs) that are validated by Federal Information Processing Standards (FIPS) 140-2 to protect keys



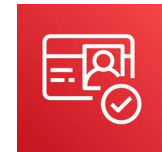
AWS Key Management
Service (AWS KMS)



Amazon Cognito

Amazon Cognito features:

- **Adds user sign-up, sign-in, and access control to your web and mobile applications.**
- Scales to millions of users.
- Supports sign-in with social identity providers, such as Facebook, Google, and Amazon; and enterprise identity providers, such as Microsoft Active Directory via Security Assertion Markup Language (SAML) 2.0.



Amazon Cognito



AWS Shield

- **AWS Shield** features:
 - Is a managed distributed denial of service (DDoS) protection service
 - Safeguards applications running on AWS
 - Provides always-on detection and automatic inline mitigations
 - *AWS Shield Standard* enabled for at no additional cost. *AWS Shield Advanced* is an optional paid service.
- Use it to **minimize application downtime and latency.**



AWS Shield



Section 5: Securing data on AWS

Module 4: AWS Cloud Security



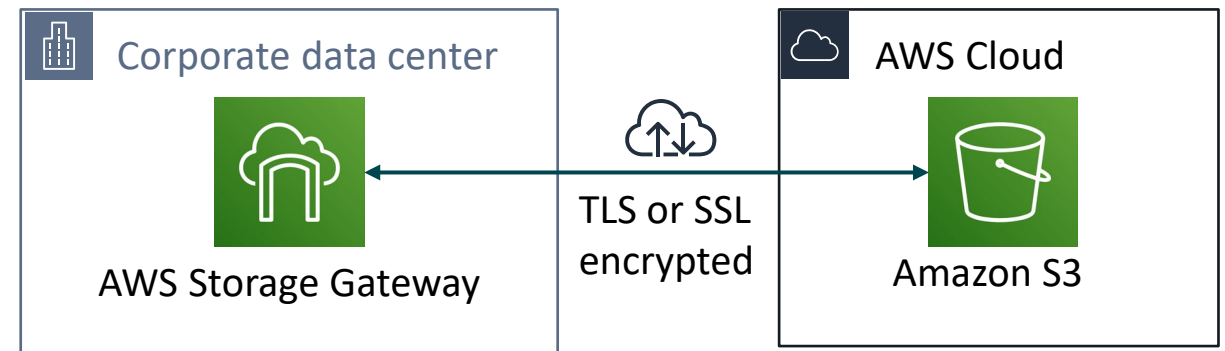
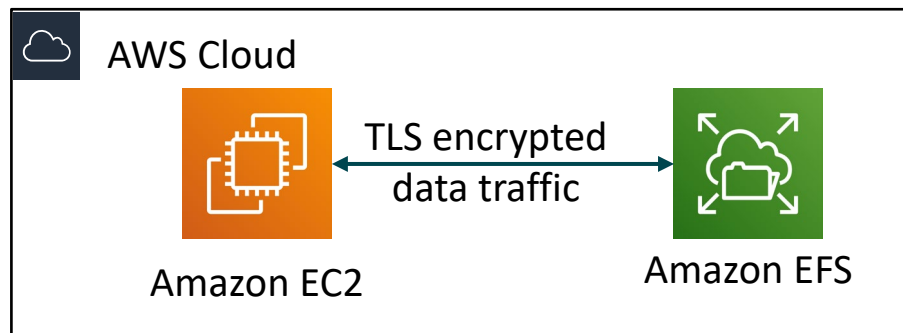
Encryption of data *at rest*

- **Encryption** encodes data with a **secret key**, which makes it unreadable
 - Only those who have the secret key can decode the data
 - **AWS KMS** can manage your secret keys
- AWS supports encryption of **data at rest**
 - Data at rest = Data stored physically (on disk or on tape)
 - You can encrypt data stored in any service that is supported by AWS KMS, including:
 - Amazon S3
 - Amazon EBS
 - Amazon Elastic File System (Amazon EFS)
 - Amazon RDS managed databases



Encryption of data *in transit*

- Encryption of **data in transit** (data moving across a network)
 - **Transport Layer Security (TLS)**—formerly SSL—is an open standard protocol
 - **AWS Certificate Manager** provides a way to manage, deploy, and renew TLS or SSL certificates
- Secure HTTP (HTTPS) creates a secure tunnel
 - Uses TLS or SSL for the bidirectional exchange of data
- **AWS services support data in transit encryption.**
 - Two examples:





Securing Amazon S3 buckets and objects

- Newly created S3 buckets and objects are **private** and **protected** by default.
- When use cases require sharing data objects on Amazon S3 –
 - It is essential to manage and control the data access.
 - Follow the **permissions that follow the principle of least privilege** and consider using Amazon S3 encryption.
- Tools and options for controlling access to S3 data include –
 - [Amazon S3 Block Public Access](#) feature: Simple to use.
 - IAM policies: A good option when the user can authenticate using IAM.
 - [Bucket policies](#)
 - [Access control lists](#) (ACLs): A legacy access control mechanism.
 - [AWS Trusted Advisor](#) bucket permission check: A free feature.



Section 6: Working to ensure compliance

Module 4: AWS Cloud Security

AWS compliance programs

- Customers are subject to many different security and compliance regulations and requirements.
- **AWS engages with certifying bodies and independent auditors to provide customers with detailed information about the policies, processes, and controls that are established and operated by AWS.**
- Compliance programs can be broadly categorized –
 - **Certifications and attestations**
 - Assessed by a third-party, independent auditor
 - Examples: [ISO 27001](#), [27017](#), [27018](#), and [ISO/IEC 9001](#)
 - **Laws, regulations, and privacy**
 - AWS provides security features and legal agreements to support compliance
 - Examples: EU [General Data Protection Regulation \(GDPR\)](#), HIPAA
 - **Alignments and frameworks**
 - Industry- or function-specific security or compliance requirements
 - Examples: Center for Internet Security (CIS), EU-US Privacy Shield certified

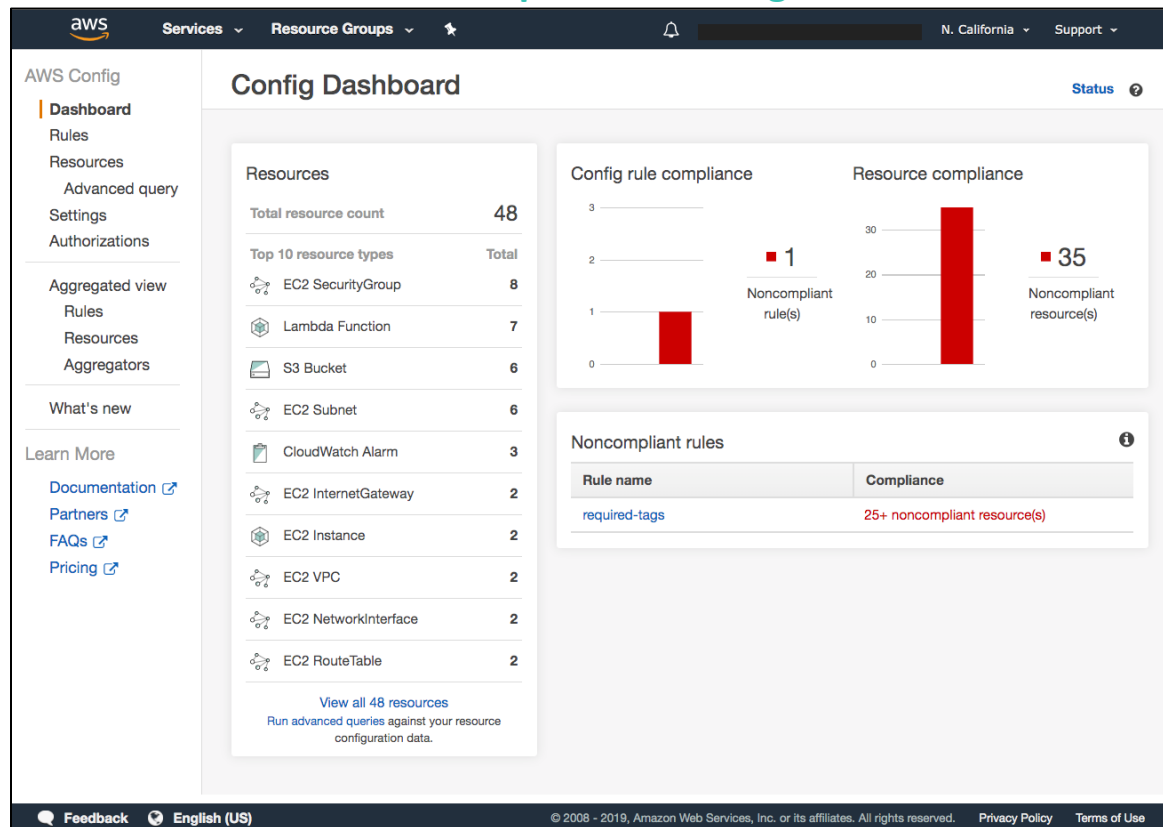


AWS Config



AWS Config

Example AWS Config Dashboard view



- Assess, audit, and evaluate the configurations of AWS resources.
- Use for continuous monitoring of configurations.
- Automatically evaluate *recorded* configurations versus *desired* configurations.
- Review configuration changes.
- View detailed configuration histories.
- Simplify compliance auditing and security analysis.



AWS Artifact



AWS Artifact

- **Is a resource for compliance-related information**
- Provide access to security and compliance reports, and select online agreements
- Can access example downloads:
 - AWS ISO certifications
 - Payment Card Industry (PCI) and Service Organization Control (SOC) reports
- Access AWS Artifact directly from the AWS Management Console
 - Under **Security, Identify & Compliance**, click **Artifact**.

Section 6 key takeaways



- **AWS security compliance programs** provide information about the policies, processes, and controls that are established and operated by AWS.
- **AWS Config** is used to assess, audit, and evaluate the configurations of AWS resources.
- **AWS Artifact** provides access to security and compliance reports.



Module wrap-up

Module 4: AWS Cloud Security





Module summary

In summary, in this module you learned how to:

- Recognize the shared responsibility model
- Identify the responsibility of the customer and AWS
- Recognize IAM users, groups, and roles
- Describe different types of security credentials in IAM
- Identify the steps to securing a new AWS account
- Explore IAM users and groups
- Recognize how to secure AWS data
- Recognize AWS compliance programs

Complete the knowledge check





Sample exam question

Which of the following is AWS's responsibility under the AWS shared responsibility model?

Choice	Response
A	Configuring third-party applications
B	Maintaining physical hardware
C	Securing application access and data
D	Managing custom Amazon Machine Images (AMIs)



Additional resources

- AWS Cloud Security: <https://aws.amazon.com/security/>
- AWS Security Resources: https://aws.amazon.com/security/security-learning/?cards-top.sort-by=item.additionalFields.sortDate&cards-top.sort-order=desc&awsf.Types=*all
- AWS Security Blog: <https://aws.amazon.com/blogs/security/>
- Security Bulletins : https://aws.amazon.com/security/security-bulletins/?card-body.sort-by=item.additionalFields.bulletinId&card-body.sort-order=desc&awsf.bulletins-flag=*all&awsf.bulletins-year=*all
- Vulnerability and Penetration testing: <https://aws.amazon.com/security/penetration-testing/>
- AWS Well-Architected Framework – Security pillar: <https://d1.awsstatic.com/whitepapers/architecture/AWS-Security-Pillar.pdf>
- AWS documentation - IAM Best Practices: <https://docs.aws.amazon.com/IAM/latest/UserGuide/best-practices.html>



Thank you



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